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Engineering in Computer Science

DELIVERABLES OF THE THIRD MODULE
Omnivest Project

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ABSTRACT

This document presents the outcomes of the third development module of the Omnivest project, covering Sprints 11 to 15. This module focused on integrating predictive intelligence, scalable infrastructure, and business validation into the Omnivest platform. Key deliverables include the deployment architecture on AWS, the creation of a robust prediction service, a full transition from regex-based to tokenization-based data extraction, enhancements to the JSON data model, and the development of Omnivest's business and financial plan. Together, these deliverables represent a crucial step in evolving Omnivest from a technical prototype into a scalable, investment-ready AI platform.

INTRODUCTION

The Omnivest project is an AI-driven financial analysis platform aimed at democratizing access to intelligent investment insights. Following the previous modules, which established system foundations and implemented secure full-stack functionality, Module 3 focuses on cloud deployment, advanced machine learning capabilities, data pipeline optimization, and financial planning. It spans Sprints 11 through 15, culminating in a mature, scalable, and business-aligned version of the system.

THIRD MODULE DELIVERABLES

Sprint 11 – AWS Project Architecture

Sprint 11 focused on defining a secure, scalable, and serverless-friendly architecture for the Omnivest platform on Amazon Web Services (AWS). The infrastructure integrates Amazon Cognito for authentication, AWS WAF for threat protection, CloudFront + S3 for static hosting and content delivery, Lambda and EC2 for compute and business logic, and DocumentDB and RDS for data persistence. This architecture ensures reliability, modularity, and high performance for production deployment.

Sprint 12 – Prediction Service and Model Development

Sprint 12 delivered a structured prediction service that bridges company financial data and the AI prediction engine. It validates inputs, prepares datasets, and communicates with the predictive model. The underlying machine learning model was refined with feature engineering, multicollinearity control, normalization, and automated comparison using PyCaret. This sprint also improved reproducibility, reliability, and interpretability of financial predictions.

Sprint 13 – Prediction API and Data Extraction Revamp

Sprint 13 implemented a FastAPI-based backend service for real-time predictions, connecting to the trained model and returning results in JSON format. It included robust error handling and data cleaning logic, enabling scalable financial forecasting through REST endpoints. Additionally, the data extraction pipeline was overhauled to replace fragile regex-based parsing with a phrase + tokenization system, dramatically improving maintainability, readability, and robustness against text variations in financial PDFs.

Sprint 14 – Company Data JSON and Model Documentation

Sprint 14 standardized the internal JSON structure (`companyData.json`) that stores company metrics by year. The schema defines consistent metadata, sector classification, yearly metrics, and financial indicators. It also introduced documentation for how the predictive model consumes this data — including lag and year-over-year features — to ensure clarity, scalability, and future-proofing of the dataset for continued model training and evaluation.

Sprint 15 – Business and Financial Planning

Sprint 15 marked the transition from a purely technical initiative to a market-oriented product. The team developed Omnivest's complete business plan, including executive summary, value proposition, market analysis, financial model, and funding strategy. It also included a comprehensive 12-month financial projection, estimating operating costs and evaluating different funding paths. This sprint connected technical feasibility with commercial sustainability, preparing Omnivest for future investment and scaling.

CONCLUSION

The third module of the Omnivest project solidified the platform's transition from a prototype to a scalable, AI-enabled, and market-ready system. Through the integration of AWS infrastructure, predictive intelligence, data standardization, and business validation, the team established a strong foundation for Omnivest's upcoming production deployment. This phase not only enhanced technical robustness but also aligned the project with real-world financial and operational goals.

REFERENCES

OMNIVEST PUBLIC REPORT – Instituto de Tecnologia e Liderança. Omnivest Methodology Documentation, 2025. AWS Documentation – Amazon Web Services, 2025. PyCaret Official Documentation – PyCaret.org, 2025. FastAPI Documentation – fastapi.tiangolo.com, 2025.